TUGAS APLIKASI KOMPUTER DAN BASIS DATA
JLIST, JTABBEDPANE, DAN JTABLE

TUGAS MINGGUAN

Disusun untuk memenuhi tugas mingguan
Aplikasi Komputer dan Basis Data

oleh :

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PROGRAM STUDI D4 TEKNIK TELEKOMUNIKASI

JURUSAN TEKNIK ELEKTRO

POLITEKNIK NEGERI BANDUNG

2024

1. Cobalah contoh program JList, JTabbedPane dan JTable yang telah dijelaskan.

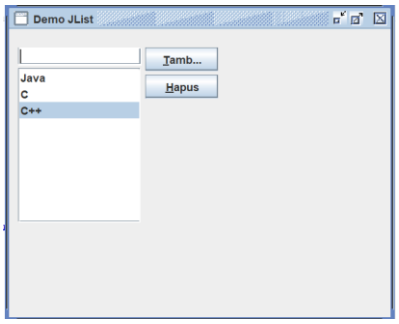
   Jawab:

Program JList:

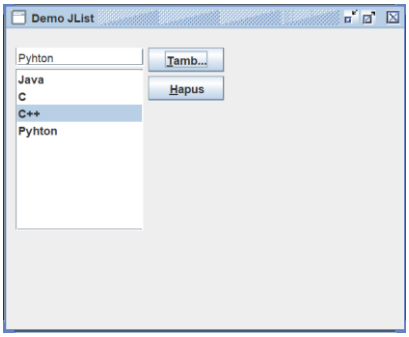
```
1  package Teori9;
2
3  import java.awt.*;
4  import java.awt.event.*;
5  import javax.swing.*;
6
7  class DemoList implements ActionListener {
8
9      public JFrame frame;
10     public JList list;
11     public JButton btnTambah, btnHapus;
12     public JTextField tfItem;
13     public JScrollPane scroller;
14
15     public DemoList() {
16         tfItem = new JTextField(12);
17         tfItem.setLocation(10,20);
18         tfItem.setSize(tfItem.getPreferredSize());
19
20         DefaultListModel lm;
21         lm = new DefaultListModel();
22         lm.addElement("Java");
23         lm.addElement("C");
24         lm.addElement("C++");
25
26         list = new JList(lm);
27         list.setSelectionMode(ListSelectionModel.SINGLE_SELECTION);
28
29         scroller = new JScrollPane(list);
30         scroller.setLocation(10,43);
31         scroller.setSize(new Dimension(135,170));
32
33         btnTambah = new JButton("Tambah");
34         btnTambah.setLocation(150,20);
35         btnTambah.setSize(btnTambah.getPreferredSize());
36         btnTambah.addActionListener(this);
37         btnTambah.setMnemonic('T');
38
39         btnHapus = new JButton("Hapus");
40         btnHapus.setLocation(150,50);
41         btnHapus.setSize(btnTambah.getPreferredSize());
42         btnHapus.addActionListener(this);
43         btnHapus.setMnemonic('H');
44     }
45
46     public void createAndSHowGUI() {
47         JFrame.setDefaultLookAndFeelDecorated(true);
48         frame = new JFrame("Demo JList");
49         frame.setLayout(null);
50         frame.setDefaultCloseOperation(JFrame.EXIT_ON_CLOSE);
51
52         frame.getContentPane().add(tfItem);
53         frame.getContentPane().add(scroller);
54         frame.getContentPane().add(btnTambah);
55         frame.getContentPane().add(btnHapus);
56
57         frame.setBounds(0,0,430,350);
58         frame.setLocationRelativeTo(null);
59         frame.setVisible(true);
60     }
61
62     public void actionPerformed(ActionEvent event) {
63         if(event.getSource() == btnTambah) {
```

Hasil Output:

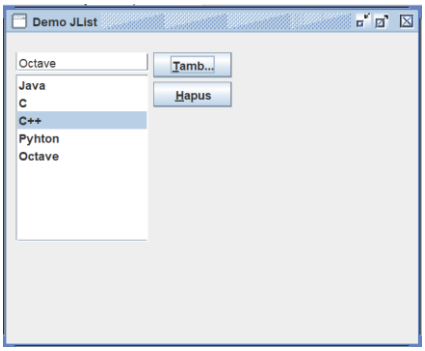
Tampilan Awal:



Penambahan Python:



Penambahan Octave:



```
            String string = tfItem.getText().trim();
            if(!string.equals("")) {
                ((DefaultListModel)list.getModel()).addElement(string);
            }
            if(list.getModel().getSize() == 1) {
                btnHapus.setEnabled(true);
            }
        } else {
            int index = list.getSelectedIndex();
            if (index == -1) {
                JOptionPane.showMessageDialog(frame,
                        "Anda harus memilih salah sati item" +
                "terlebih dahulu");
                return;
            }
            ((DefaultListModel)list.getModel()).remove(index);
            int size = list.getModel().getSize();

            if(size == 0) {
                btnHapus.setEnabled(false);
            } else {
                if(index == list.getModel().getSize()) {
                    index--;
                }
                list.setSelectedIndex(index);
                list.ensureIndexIsVisible(index);
            }
        }
    }

    public static void main (String[] args) {
        javax.swing.SwingUtilities.invokeLater(new Runnable() {
            public void run() {
                DemoList app = new DemoList();
                app.createAndSHowGUI();
            }
        });
    }
}
```

Program JTabbedPane:

```
package Teori9;

import java.awt.*;
import java.awt.event.*;
import javax.swing.*;
import java.io.Serializable;

class Page1 extends JPanel implements ActionListener, Serializable {
    private static final long serialVersionUID = 1L;
    private JButton btn;

    public Page1() {
        btn = new JButton("Info Page #1");
        btn.addActionListener(this);
        btn.setMnemonic('1');
        add(btn);
    }

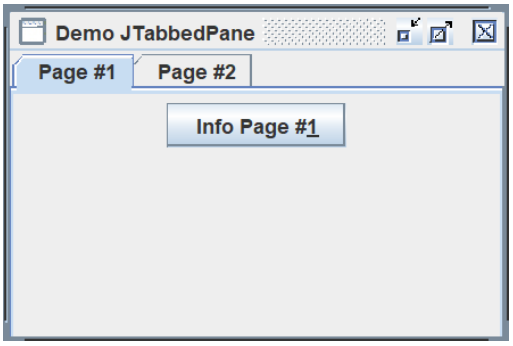
    public void actionPerformed(ActionEvent event) {
        if(event.getSource() == btn) {
            JOptionPane.showMessageDialog(null, "ini adalah page ke-1");
        }
    }
}

class Page2 extends JPanel {
    private JTextField tf;

    public Page2() {
        tf = new JTextField("ini adalah contoh page kedua", 20);
        add(tf);
    }
```

Hasil Output:

Tampilan pada Page #1:



Jika Info Page #1 ditekan:



```
}

class DemoTabbedPane {
    public JTabbedPane tp;

    public DemoTabbedPane() {
        ImageIcon icon = new ImageIcon("/image/icon.jpg");

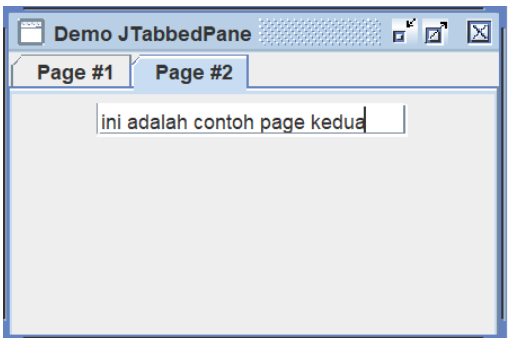
        tp = new JTabbedPane();
        tp.addTab("Page #1", icon, new Page1(), "Page pertama");
        tp.addTab("Page #2", icon, new Page2(), "Page kedua");

        tp.setLocation(0, 0);
        tp.setSize(new Dimension(290, 190));
    }

    public void createAndShowGUI() {
        JFrame.setDefaultLookAndFeelDecorated(true);
        JFrame frame = new JFrame("Demo JTabbedPane");
        frame.setLayout(null);
        frame.setDefaultCloseOperation(JFrame.EXIT_ON_CLOSE);
        frame.getContentPane().add(tp);
        frame.setBounds(0, 0, 300, 200);
        frame.setLocationRelativeTo(null);
        frame.setVisible(true);
    }

    public static void main(String[] args) {
        javax.swing.SwingUtilities.invokeLater(new Runnable() {
            public void run() {
                DemoTabbedPane app = new DemoTabbedPane();
                app.createAndShowGUI();
            }
        });
    }
}
```

Tampilan pada Page #2:



Program JTabble:

```
package Teori9;

import java.awt.*;
import javax.swing.*;

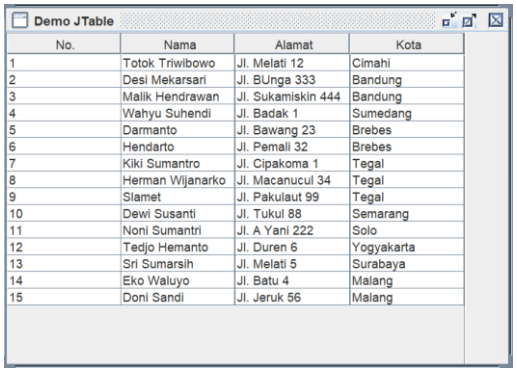
class DemoTable {
    public JTable table;
    public JScrollPane scroller;

    public DemoTable() {
        final String[] judulKolom =
            {"No.", "Nama", "Alamat", "Kota"};

        final Object[][] data = {
            {"1", "Totok Triwibowo", "Jl. Melati 12", "Cimahi"},
            {"2", "Desi Mekarsari", "Jl. BUnga 333", "Bandung"},
            {"3", "Malik Hendrawan", "Jl. Sukamiskin 444", "Bandung"},
            {"4", "Wahyu Suhendi", "Jl. Badak 1", "Sumedang"},
            {"5", "Darmanto", "Jl. Bawang 23", "Brebes"},
            {"6", "Hendarto", "Jl. Pemali 32", "Brebes"},
            {"7", "Kiki Sumantro", "Jl. Cipakoma 1", "Tegal"},
            {"8", "Herman Wijanarko", "Jl. Macanucul 34", "Tegal"},
            {"9", "Slamet", "Jl. Pakulaut 99", "Tegal"},
            {"10", "Dewi Susanti", "Jl. Tukul 88", "Semarang"},
            {"11", "Noni Sumantri", "Jl. A Yani 222", "Solo"},
            {"12", "Tedjo Hemanto", "Jl. Duren 6", "Yogyakarta"},
            {"13", "Sri Sumarsih", "Jl. Melati 5", "Surabaya"},
            {"14", "Eko Waluyo", "Jl. Batu 4", "Malang"},
            {"15", "Doni Sandi", "Jl. Jeruk 56", "Malang"}
        };

        table = new JTable(data, judulKolom);
```

Hasil Output:



No.	Nama	Alamat	Kota
1	Totok Triwibowo	Jl. Melati 12	Cimahi
2	Desi Mekarsari	Jl. BUnga 333	Bandung
3	Malik Hendrawan	Jl. Sukamiskin 444	Bandung
4	Wahyu Suhendi	Jl. Badak 1	Sumedang
5	Darmanto	Jl. Bawang 23	Brebes
6	Hendarto	Jl. Pemali 32	Brebes
7	Kiki Sumantro	Jl. Cipakoma 1	Tegal
8	Herman Wijanarko	Jl. Macanucul 34	Tegal
9	Slamet	Jl. Pakulaut 99	Tegal
10	Dewi Susanti	Jl. Tukul 88	Semarang
11	Noni Sumantri	Jl. A Yani 222	Solo
12	Tedjo Hemanto	Jl. Duren 6	Yogyakarta
13	Sri Sumarsih	Jl. Melati 5	Surabaya
14	Eko Waluyo	Jl. Batu 4	Malang
15	Doni Sandi	Jl. Jeruk 56	Malang

```
33
34          scroller = new JScrollPane(table,
35            ScrollPaneConstants.VERTICAL_SCROLLBAR_AS_NEEDED,
36            ScrollPaneConstants.HORIZONTAL_SCROLLBAR_AS_NEEDED);
37          scroller.setLocation(0,0);
38          scroller.setSize(new Dimension(440,330));
39      }
40
41      public void createAndShowGUI() {
42          JFrame.setDefaultLookAndFeelDecorated(true);
43          JFrame frame = new JFrame("Demo JTable");
44          frame.setLayout(null);
45          frame.setDefaultCloseOperation(JFrame.EXIT_ON_CLOSE);
46
47          frame.getContentPane().add(scroller);
48          frame.setBounds(0,0,490,350);
49          frame.setLocationRelativeTo(null);
50          frame.setVisible(true);
51      }
52
53      public static void main (String[] args) {
54          javax.swing.SwingUtilities.invokeLater(new Runnable() {
55              public void run() {
56                  DemoTable app = new DemoTable();
57                  app.createAndShowGUI();
58              }
59          });
60      }
61  }
```

2. Dari program JTabbedPane dapat dikembangkan, misal ; membuat program JTabbedPane yang terdiri dari 4 page, dimana page1 berisi form signup, page2 berisi form menu, page3 berisi form pemesanan, page4 berisi form data transaksi.

   Jawab:

Program	Hasil Output

```
1  package Teori9;
2
3  import java.awt.*;
4  import java.awt.event.*;
5  import javax.swing.*;
6  import java.io.Serializable;
7
8  class SignUpForm extends JPanel implements ActionListener, Serializable {
9      private static final long serialVersionUID = 1L;
10     private JLabel nameLabel, emailLabel, passwordLabel;
11     private JTextField nameField, emailField;
12     private JPasswordField passwordField;
13     private JButton signUpButton;
14
15     public SignUpForm() {
16         setLayout(new GridLayout(4, 2));
17
18         nameLabel = new JLabel("Name:");
19         nameField = new JTextField();
20         add(nameLabel);
21         add(nameField);
22
23         emailLabel = new JLabel("Email:");
24         emailField = new JTextField();
25         add(emailLabel);
26         add(emailField);
27
28         passwordLabel = new JLabel("Password:");
29         passwordField = new JPasswordField();
30         add(passwordLabel);
31         add(passwordField);
```

Tampilan Page Sign Up:

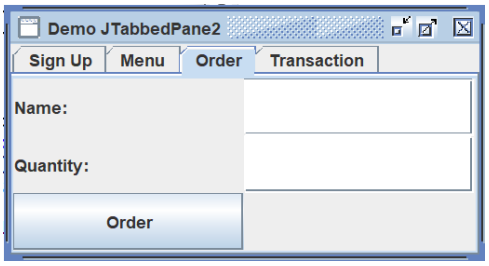


Tampilan Page Menu:

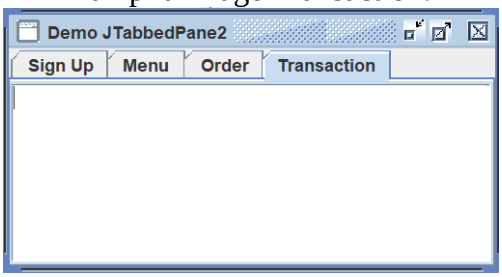


```
32
33          signUpButton = new JButton("Sign Up");
34          add(signUpButton);
35      }
36
37      public void actionPerformed(ActionEvent event) {
38      }
39  }
40
41  class MenuForm extends JPanel implements Serializable {
42      private static final long serialVersionUID = 1L;
43      private JButton menuButton1, menuButton2, menuButton3;
44
45      public MenuForm() {
46          setLayout(new GridLayout(3, 1));
47
48          menuButton1 = new JButton("Menu 1");
49          menuButton2 = new JButton("Menu 2");
50          menuButton3 = new JButton("Menu 3");
51
52          add(menuButton1);
53          add(menuButton2);
54          add(menuButton3);
55      }
56  }
57
58  class OrderForm extends JPanel implements Serializable {
59      private static final long serialVersionUID = 1L;
60      private JLabel nameLabel, quantityLabel;
61      private JTextField nameField, quantityField;
62      private JButton orderButton;
63
64      public OrderForm() {
65          setLayout(new GridLayout(3, 2));
66
67          nameLabel = new JLabel("Name:");
68          nameField = new JTextField();
69          add(nameLabel);
70          add(nameField);
71
72          quantityLabel = new JLabel("Quantity:");
73          quantityField = new JTextField();
74          add(quantityLabel);
75          add(quantityField);
76
77          orderButton = new JButton("Order");
78          add(orderButton);
79      }
80  }
81
82  class TransactionForm extends JPanel implements Serializable {
83      private static final long serialVersionUID = 1L;
84      private JTextArea transactionData;
85
86      public TransactionForm() {
87          setLayout(new BorderLayout());
88
89          transactionData = new JTextArea();
90          JScrollPane scrollPane = new JScrollPane(transactionData);
91          add(scrollPane, BorderLayout.CENTER);
92      }
93  }
94
95  class DemoTabbedPane2 {
```

Tampilan Page Order:



Tampilan Page Transaction:



```
    public JTabbedPane tp;

    public DemoTabbedPane2() {
        tp = new JTabbedPane();
        tp.addTab("Sign Up", new SignUpForm());
        tp.addTab("Menu", new MenuForm());
        tp.addTab("Order", new OrderForm());
        tp.addTab("Transaction", new TransactionForm());
    }

    public void createAndShowGUI() {
        JFrame.setDefaultLookAndFeelDecorated(true);
        JFrame frame = new JFrame("Demo JTabbedPane2");
        frame.setDefaultCloseOperation(JFrame.EXIT_ON_CLOSE);
        frame.getContentPane().add(tp);
        frame.pack();
        frame.setLocationRelativeTo(null);
        frame.setVisible(true);
    }

    public static void main(String[] args) {
        javax.swing.SwingUtilities.invokeLater(new Runnable() {
            public void run() {
                DemoTabbedPane2 app = new DemoTabbedPane2();
                app.createAndShowGUI();
            }
        });
    }
}
```